

8 Conclusion

We introduced post-intervention recovery as a diagnostic for testing whether SAE feature interventions form complete behavioral bottlenecks. Rather than asking whether a selected feature set can change behavior when clamped, our setting asks whether the suppressed behavior is actually eliminated once the clamp remains active. Across latent-level, output-level, circuit-level, and refusal settings, we find that suppressed behaviors can often be recovered from the defended residual state while the targeted SAE features remain close to their defended values. The refusal attribution analysis further shows that recovery is not primarily explained by reopening the clamped features or shifting into a small set of alternative SAE latents, but by behaviorally sufficient information in the SAE-unexplained residual component. These findings separate causal usefulness from intervention completeness: SAE features can provide valuable local handles for analysis and control, but successful feature suppression should not be treated as proof of behavioral elimination. SAE-based defenses should therefore be evaluated not only by their immediate suppression effect, but also by the robustness of the defended state to constrained post-intervention recovery.

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